

Sean Tran

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Education

The University of Texas at Dallas	B.S. in Physics (Expected Dec 2026) — GPA: 3.3	2024 – Present
	B.S. in Computer Science	2019 – 2024

Work Experience

Undergraduate Researcher — Computational Astrophysics <i>University of Texas at Dallas – Richardson, TX</i>	May 2025 – June 2026
- Employed statistical techniques to identify the source location of gravitational-wave signals by correlating arrival times with their electromagnetic counterpart - Developed and optimized Python-based simulations for gravitational lensing in a Linux environment, executing large-scale parameter sweeps across high-dimensional spaces to improve computational efficiency - Implemented numerical ray-tracing algorithms for modeling optical systems under varying condition, extending models toward wave-based effects and using image-based visualization to interpret system behavior - Presented data-driven results at UTD SPUR 2025; selected for research award - Tools Used: <i>Python (NumPy, SciPy, Astropy, Matplotlib, Plotly), MATLAB, Linux, Bash, Jupyter, Git, Glafic</i>	
Restaurant Associate — Family-Owned Business <i>Golden Deli – Addison, TX</i>	June 2016 – Present
Managed operations including transactions, inventory control, and equipment maintenance while troubleshooting issues	

Projects and Builds

Encrypted Password Manager – Secure Storage of Sensitive Data
- Developed a modern password manager backed by PostgreSQL with encrypted credential storage, account management, configurable security settings, and relational data integrity through foreign-key constraints and cascading deletion - Implemented AES-256 encryption with Argon2-based key derivation and secure master-password rotation, using atomic database transactions to maintain data integrity and safely re-encrypt stored credentials - Tools Used: <i>Python, PostgreSQL, Docker, Git</i>
Modular Report Generation Tool – Senior Capstone Project (CS)
- Led a team of 4 in developing a program to automate audit report generation for ECS Graduate Advising at UTD, extensively increasing initial process efficiency by 250% and future modifications by an even larger amount - Designed and implemented a modular architecture utilizing object-oriented principles, to divide the project into four crucial components, allowing for efficient collaboration, future scalability and streamlining final integration process - Tools Used: <i>C#, .NET, Powershell, AWS</i>
Laser Diode Spectroscopy — Hyperfine Transitions of Rubidium
- Diagnosed and corrected a systematic measurement error in a laser setup caused by unintended optical paths in the Michelson interferometer, reducing error from an order-of-magnitude deviation to $< 5\%$ - Developed a fringe-based frequency calibration to extract GHz-scale spectral features and determine hyperfine peak separations consistent with known atomic transitions
Optical Schlieren Imaging System — Density Gradients in Gas Propulsion
- Designed and built an optical Schlieren imaging system to capture refractive index variations in gas flow, enabling visualization of otherwise undetectable flow structures - Aligned and optimized optical components (mirrors, a collimated light source, and knife-edge cut-off) to optimize system sensitivity, spatial resolution, and image contrast

Skills

Programming: Java, Python, C++/C/C#, Bash, SQL, HTML/CSS,
Software: Linux (Ubuntu/WSL), Git, MATLAB, Docker, Jupyter, MS Excel, AWS, Arduino IDE

Work Eligibility: Eligible to work full time in the U.S. with no restrictions